

Solution
FORCE AND PRESSURE -04
Class 08 - Science
Section A

1. (c) A is true but R is false.
Explanation: An apple falls from a tree because of gravitational attraction between the earth and the apple. So, (A) is the correct statement.
Gravitational force is a non-contact force. For the action of gravitational force, contact with an object is not required. So, (R) is an incorrect statement.
2. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: When a stone is thrown vertically upward, after getting a certain height' it comes back to the earth because of gravitation force. So, (A) is the correct statement.
When the stone is released, it falls down. Once the stone is released, the forces acting on the stone are the gravitational force and an upward force opposing the gravitational force. If the medium in which the stone is released is air, frictional force due to air also acts on the stone opposing its motion. So, (R) is the correct statement.
So, amongst these correct options is (B) both assertion and reason are correct but reason is not the correct explanation for assertion.
3. (a) Both A and R are true and R is the correct explanation of A.
Explanation: $\text{Pressure} = \frac{\text{force}}{\text{area}}$
A nail has a pointed end so that the surface area is less, when the surface area is less the pressure increases which makes it easy for the nail to be driven into the wall or where ever it may be inserted.
So, Both A and R are true and R is the correct explanation of A.
4. (d) A is false but R is true.
Explanation: In space, outside of the Earth's atmosphere, there are almost no molecules in the gas that surrounds the International Space Station. The absence of molecules means that the air pressure is extremely low – in fact, it is almost zero, so it is nearly a perfect vacuum. So, (A) is an incorrect statement.
The suits protect astronauts from getting too hot or cold. Spacesuits also give astronauts oxygen to breathe while they are working in space. So, (R) is the correct statement.
5. (c) A is true but R is false.
Explanation: Sea divers wear special suits to prevent their bodies from being crushed by the tremendous pressure of water exerted on all sides. So, (A) is the correct statement.
Gravity is one major force that creates tides. This means gravity affects the sea also. So, (R) is an incorrect statement.
6. (c) A is true but R is false.
Explanation: The motion imparted to objects was due to the action of a force. So, (A) is false.
In science, a push or a pull on an object is called a force. Is the correct definition of force. So, (R) is true.
7. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Magnetic forces are non-contact forces; they pull or push on objects without touching them. Magnets are only attracted to a few 'magnetic' metals and not all matter.
Like poles of the magnets i.e. north-north and south-south repel each other and unlike poles of the magnets i.e. north-south and south-north attract each other.
So, Both A and R are true and R is the correct explanation of A.
8. (a) Both A and R are true and R is the correct explanation of A.
Explanation: When we stop pedaling, the bicycle begins to slow down. This is again because of the friction forces acting

opposite to the direction of motion. In order to keep the bicycle moving, we have to start pedaling again.

So, Both A and R are true and R is the correct explanation of A.

9. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: This force is caused by the action of muscles in our body. The force resulting due to the action of muscles is known as the muscular force.

So, Both A and R are true and R is the correct explanation of A.

10. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: While kicking a ball, the footballer's leg will experience the same magnitude of force by which it kicked on the ball but, in the opposite direction.

The goalkeeper applies a force on the ball in the opposite direction to that of its motion and thus stops the ball.

So, Both A and R are true and R is the correct explanation of A.

Section B

11. State True or False:

- (i) **(a)** True

Explanation: True

- (ii) **(a)** True

Explanation: True

- (iii) **(a)** True

Explanation: True

- (iv) **(a)** True

Explanation: True

- (v) **(a)** True

Explanation: True

- (vi) **(b)** False

Explanation: False

- (vii) **(a)** True

Explanation: True

- (viii) **(a)** True

Explanation: True

- (ix) **(a)** True

Explanation: True

- (x) **(b)** False

Explanation: False

- (xi) **(b)** False

Explanation: False

- (xii) **(b)** False

Explanation: False

- (xiii) **(b)** False

Explanation: False

- (xiv) **(a)** True

Explanation: True

- (xv) **(a)** True

Explanation: True

- (xvi) **(b)** False

Explanation: False

- (xvii) **(b)** False

Explanation: False

(xviii)(a) True

Explanation: True

(xix) (a) True

Explanation: True

Section C

12. Fill in the blanks:

- (i) 1. Repels
- (ii) 1. Pressure
- (iii) 1. Push, pull
- (iv) 1. Apply force (lift)
- (v) 1. force
- (vi) 1. Gravity, friction.
- (vii) 1. Shape
- (viii) 1. friction
- (ix) 1. contact
- (x) 1. Muscular
- (xi) 1. muscular force
- (xii) 1. pressure
- (xiii) 1. contact
- (xiv) 1. Pull/push
- (xv) 1. Attracts
- (xvi) 1. Interaction

Section D

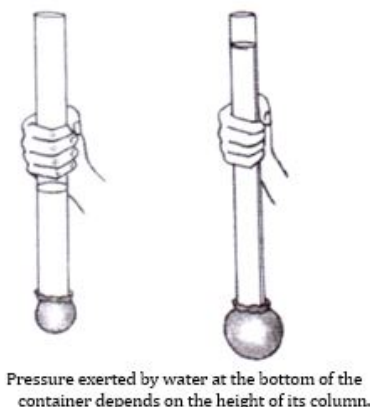
- 13. a. - r
b. - s
c. - p
d. - q
- 14. a. - r
b. - s
c. - p
d. - q
- 15. a. - r
b. - p
c. - s
d. - q
- 16. a. - q
b. - s
c. - p
d. - r
- 17. a. - r
b. - p
c. - s
d. - q

18.	Column A	Column B
	(i) Force	(c) Push or pull
	(ii) Contact force	(d) Muscular force
	(iii) Magnitude	(e) Strength of force
	(iv) Force of gravity	(a) Non-contact force

	(v) Pressure	(b) Force on a unit area
19.	Column A	Column B
	(i) Magnet	(a) Non-contact force
	(b) Gravity	(b) Earth
	(iii) Pressure	(a) Unit area
	(iv) Friction force	(e) Contact force
	(v) Force	(d) Pull or push

Section E

20. i. Muscular force- The force exerted by the muscles of our body is called the muscular force. Both human beings and animals can exert force with their muscles. For example, we push a cart with our muscles, child riding a bicycle is using his legs muscles to ride a cycle, and a person drawing water from the well is using his muscular force.
- ii. Frictional force- A force that opposes the motion of an object is called a frictional force. It acts between the surface of the two objects. For example, a ball rolling on the ground slows down and then stops after some time. It is due to the frictional force acting between the ground and the surface of the ball.
21. Take a transparent glass tube or plastic pipe. Also take piece of thin sheet of a good quality rubber. Stretch the rubber sheet tightly over one end of pipe. Hold the pipe at the known as middle, keeping it in a vertical position. Pour some water in the pipe. Note the height of the water column in the pipe. Pour some more water. Observe, the bulge in rubber lay when sheet and height of water column in the pipe. Repeat this process a few more times. You observe that as the height of water column to the increases the bulge in the rubber sheet also increases.



We conclude that:

- A liquid exerts pressure on the bottom of its container.
 - The pressure exerted by a liquid depends on the height of the liquid column.
22. A push or a pull on an object is called a force. It results in various changes in the body. Examples of forces in everyday life are like pulling, pushing, lifting, stretching, twisting and pressing any object. A force is used when we kick a ball, a force is used when we pull a cart, when we squeeze a lemon etc.
23. While walking, we apply force on the ground. The ground applies force on our foot due to which we are able to move. This is called the force of friction.
- The force of friction always acts on all the moving objects.
 - Its direction is always opposite to the direction of motion.
24. Electrostatic force is the force which might be responsible for this phenomenon because:
- The electrostatic force is the force that is produced when charge bodies come together.
 - It is given that two thermocol balls have similar charges.
 - As we know that similar charges repel each other.
 - Hence, they move away from each other due to the repulsion between them.
25. Otto von Guericke, a German Scientist invented a pump in 17th century to extract air out of a vessel. He demonstrated the force of the air pressure. He joined two hollow metallic hemispheres of 51 cm diameter each and pumped air out of them. Then he employed eight horses on each hemisphere to pull them apart. So great is the force of air pressure that the hemispheres could not be pulled apart.



Horses pulling the hemispheres.

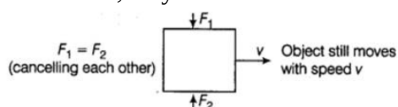
26. Take a plastic bottle. Fix a small glass tube above the bottom of the plastic bottle. We should seal the joint of glass tube with plastic bottle by using molten wax so that water does not leak from the joint. Tie a thin sheet of rubber tightly on the open end of glass tube. Now fill half of plastic bottle with water. On filling water, the rubber sheet tied to the mouth of glass tube gets stretched and bulges out. The bulging out of rubber sheet tied to the glass tube fixed in the wall of plastic bottle demonstrates that water present in plastic bottle exerts pressure on the walls of the bottle. It is the sideways pressure exerted by water which inflates the thin rubber sheet forming a bulge. If we pour more water in the plastic bottle to increase depth, we will see that the bulge in the rubber sheet increases. This indicates that the pressure exerted by water increases with increasing depth.

- i. The pressure exerted by a liquid on the walls of container increases with increasing depth.
- ii. A liquid exerts pressure on the walls of its container.



Liquid exerts pressure on the walls of the container.

27. The force exerted by an electrically charged object is called an electrostatic force.
A charged body can exert a force on an uncharged body, which makes it charged too.
For example, a plastic comb rubbed on dry hair becomes charged and can attract tiny pieces of paper.
Electrostatic force is a non contact force because it can act even at a distance.
It is like a magnetic force, it can attract of the two charged bodies carry opposite charges and they can repel if the two bodies are having the same types of charges.
28. When fruits detached from a tree fall down due to force of gravity, the objects interacting in this are fruits and earth.
- i. As soon as the fruits detached from a tree, the earth applies its force of gravity on fruit.
 - ii. The force of gravity will act towards the centre of the fruit.
 - iii. Hence, the pull of the earth causes fruits to fall down and interact with the earth.
29. The forces which can be exerted from a distance, without establishing a contact non-contact forces. Some non-contact forces are:
- (i) Magnetic force:** The force exerted by a magnet on other magnet or some magnetic substance like iron is called magnetic force. Like poles of a each other and unlike poles of a magnet attract each other without contact. called non-contact force.
- (ii) Electrostatic force:** The force exerted by a charged body on other char* uncharged body is called electrostatic force. Electrostatic force also acts making a direct contact with other charged or uncharged body. So it is also a non-contact force.
- (iii) Force of gravity:** Earth pulls every thing or body towards it. The force of attraction exerted by earth on any object is called force of gravity. This is also a non-contact force as it acts from a distance.
30. We infer about the magnitudes and direction of the forces applied that:
- i. The magnitude of forces will be equal and the same.
 - ii. The forces will act in the opposite direction.
 - iii. As a result, they cancel out each other and the net effect will become less.



31. The forces which come into play only when two objects come in contact are called contact forces.

Some contact forces are:

- (i) Muscular force:** The force resulting due to the action of muscles is called Muscular force. This force can be applied only when it is in contact with an object. It is because of this force that human beings, as well as animals are able to perform various tasks. E.g. pushing an object, lifting a bucket of water.
- (ii) Force of friction:** When a body is moving, then a force equal and direction of motion is exerted on that moving body. This

force is called force. It is also a contact force because it is exerted when two surfaces come in contact with each other. If there is no contact between two surfaces, there would be no friction.

32. The magnetic force causes the deflection because:

- i. When the key is closed, the current starts flowing in the wire.
- ii. Due to the flowing current, the magnetic field is produced around the circuit.
- iii. This exerts a magnetic force on the needle.
- iv. As a result, the needle gets deflected.

33. Force can be smaller or larger, that means it has a magnitude. Its magnitude can be expressed in Newton which is the SI unit of force.

If the two forces are applied on an object in the same direction, then the resultant force acting on the object is the sum of the two forces. For example, two boys pushing a heavy box in the same direction, their combined force is the resultant force on the box.

If the two forces applied to an object are equal but in opposite directions, then the total force is zero or nil. They are balanced or cancelled.

If the two forces applied to an object are in the opposite directions, then the resultant force is the difference of the two forces.

34. The force exerted by a magnet is called magnetic force.

It is a non-contact force as it can exert its magnetic force even from a distance. There are two types of magnetic forces:

- (i) There is a magnetic force of repulsion between the like poles of two magnets.
- (ii) There is magnetic force of attraction between the two opposite poles of a magnet.

Magnetic force can be widely used in everyday life, for example, many toys work on the principle of magnetic force, refrigerator's stickers are also made up of magnets.

35. It is much easier to burst an inflated balloon with a needle than by a finger because the pressure is inversely proportional to the area of cross-section. Hence, the smaller the area, the larger will be the pressure.

The needle has a smaller area of contact compared to the finger. Thus, the needle will exert more pressure on the balloon. Larger pressure by needle bursts an inflated balloon.

36. It is difficult to cut cloth using a pair of scissors with blunt blades because:

- i. Blunt blades (sharp) have a large surface area than sharp-edged blades.
- ii. As we know that pressure is inversely proportional to the area. Larger the area, smaller will be the pressure.
- iii. Hence, in blunt blades, the applied force produces very less pressure.
- iv. Due to low pressure, this becomes difficult to cut the cloth from blunt blades.

37. The effects of force applied on an object are as follows:

- a. A force can move a stationary body.
- b. A force can stop a moving body.
- c. A force can change the speed of the moving body.
- d. A force can change the direction of the moving body.
- e. A force can change the shape and size of the body.

38. Magnetic force in the upward direction and force of gravity in the downward direction acting on the car. Magnetic force is larger than the force of gravity because:

- i. The car is sticking to an electromagnet which is lifting the car upwards.
- ii. The car is moving upwards.

Hence, the magnetic force due to electromagnet is larger than the force of gravity.

39. Rod B will sink more because:

- i. As we know that the smaller the area, the larger the pressure on a surface for the same force applied.
- ii. Rod B has a smaller area.
- iii. Hence, it will exert more pressure on the sand.
- iv. On the other hand, rod A will exert less pressure on the sand as it has a larger area.

40. Men should follow the steps given below:

- i. If the force is applied in the direction opposite to the direction of motion, then it results in a decrease in the speed of the object.
- ii. Hence, to slow down the cart, he should apply an opposite force to the moving cart.
- iii. To balance the downward motion of the cart which is due to the force of gravity, he can start pulling the cart instead of pushing the cart.

41. Gravitational force (force of gravity) is the force that is responsible for his downward motion. Whenever any object or thing moves in a downward motion or falls towards the earth, the force of gravity takes place.
Now, he will not come down with the same speed without the parachute because the friction of air becomes less and as a result, he will come down with the fastest speed as much as possible.
42. The forces which can be exerted from a distance, without establishing a contact non-contact forces.
Some non-contact forces are:
- (i) Magnetic force:** Magnetic force is a kind of force that magnets exhibit to pull or push other magnets and metal objects like iron and nickel. If the same pole of two magnets are placed near each other they will repel, while if unlike poles are placed near each other they attract each other.
 - (ii) Electrostatic force:** Electrostatic is a type of force that is exerted by a charged body on another charged or uncharged body. It pulls lighter objects towards itself once it is charged. Electric air purifiers in factories use this concept to purify the air from its chimneys.
 - (iii) Force of gravity:** Every object in the universe exerts a force on another object. This force is called gravitational force. It is the gravitational force of various celestial bodies which keeps them in their position in the space. This is a non-contact force as it acts from a distance.